

Ontology matching, standardisation and interoperability

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Heterogeneity problem

Semantic heterogeneity

Ontology matching

Plan

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Increasing **heterogeneity** in data ecosystems (volume and complexity).

Interoperability is a key challenge in many scenarios:

- Data sharing
- Data integration
- Data translation
- Query federation
- ...

Types of Heterogeneity

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Heterogeneity of data is generally observed at three levels:

- **Semantic**: different ways of interpreting the meaning of data (role of ontologies).
- **Syntactic**: different ways of formatting the data (role of standards).
- **Structural**: different ways of storing data.

Semantic Web Standards

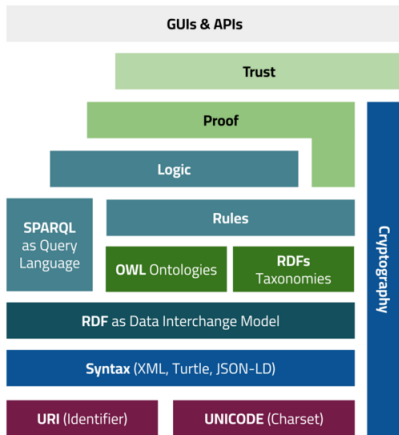
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Semantic Web is a collaborative movement led by international standards body the World Wide Web Consortium (W3C).

Standards promote common data formats on the World Wide Web.



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- Semantic heterogeneity can be partially addressed through ontologies (shared contextualisation).
- Rang of ontologies been developed to semantically represent domain concepts.
- Ontologies constructed by different designers with different views on how to model the world.

Ontology heterogeneity

Ontology differences in terms of the terminology, coverage, granularity modelling strategies, or still level of generality

Ontology matching

Task of generating a set of correspondences between different ontologies

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Kinds of heterogeneity

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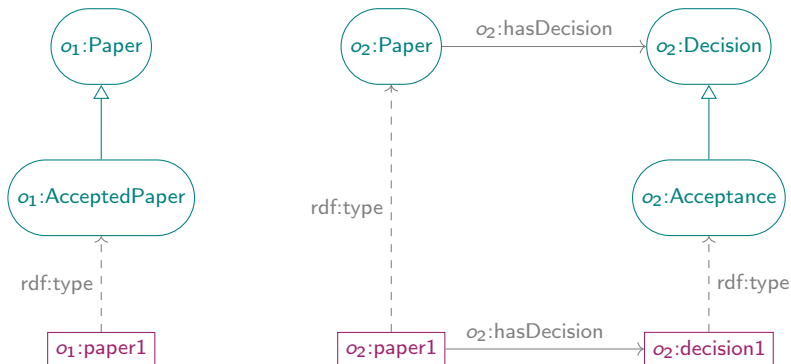
- Different languages are used for representing ontologies (XML, SKOS, RDFS, OWL)
- Different terminologies are used
 - English vs. Chinese
 - Book vs. Monograph
- Different models are used
 - Different classes: Book vs. Paperback
 - Classes vs. property: Essay vs. literarygenre
 - Classes vs. instances: one physical book as an instance vs. one work as an instance
- Different scopes and granularity are used
 - Only books vs. all cultural items vs. any product

Example

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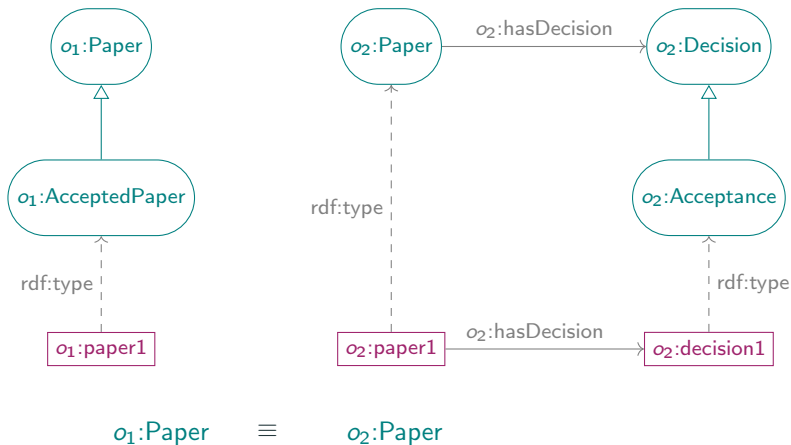


Example

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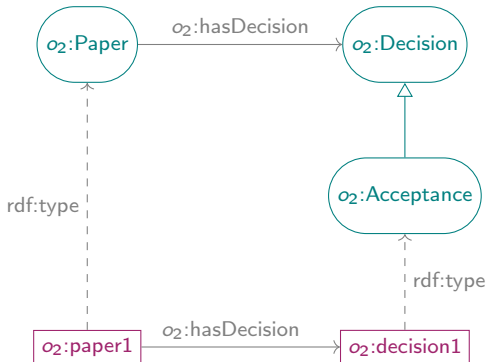
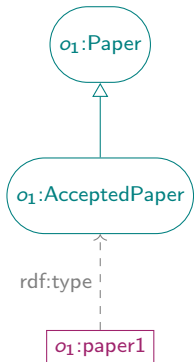


Example

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`o1:Paper` $\equiv$

`o2:Paper`

`o1:AcceptedPaper` $\equiv$

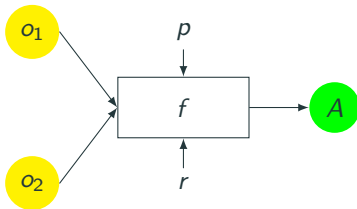
`o2:Paper` $\sqcap \exists o_2:\text{hasDecision}.o_2:\text{Acceptance}$

Ontology matching process

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Adapted from [Euzenat and Shvaiko, 2013]

A is a set of **correspondences** $\{c_1, \dots, c_n\}$, where c_i is a tuple (e_1, e_2, r)
 e_1 and e_2 are the members of the correspondence.

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 e_1 and e_2 are the members of the correspondence:

- **simple** correspondence (s:s): e_1 and e_2 are simple expressions
 $(o_1:\text{Paper}, o_2:\text{Paper}, \equiv)$
- **complex** correspondence (s:c, c:s, c:c): e_1 or/and e_2 is a complex expression
 $(o_1:\text{AcceptedPaper}, \exists o_2:\text{Paper} \sqcap o_2:\text{hasDecision}.\text{Acceptance}, \equiv)$
- r is a relation, e.g., $(\equiv, \supseteq, \sqsubseteq, \perp)$

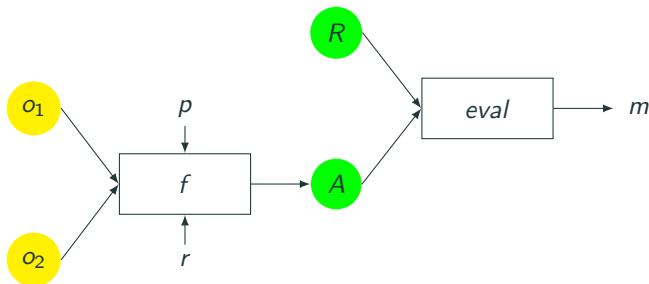
- **Input dimensions**
 - Underlying models (e.g., XML, SKOS, RDFS, OWL)
 - Schema-level vs. instance-level
- **Process dimensions**
 - Approximate vs. exact
 - Interpretation of the input
- **Output dimensions**
 - Cardinality (e.g., 1-1, 1-*)
 - Equivalence vs. diverse relations (e.g., subsumption)
 - Graded vs. absolute confidence

Ontology matching evaluation

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Adapted from [Euzenat and Shvaiko, 2013]

Evaluation dimensions and criteria:

- Tool-oriented: efficiency, scalability
- Output-oriented: compliance to a reference alignment, coherence, conservativity, suitability to the task

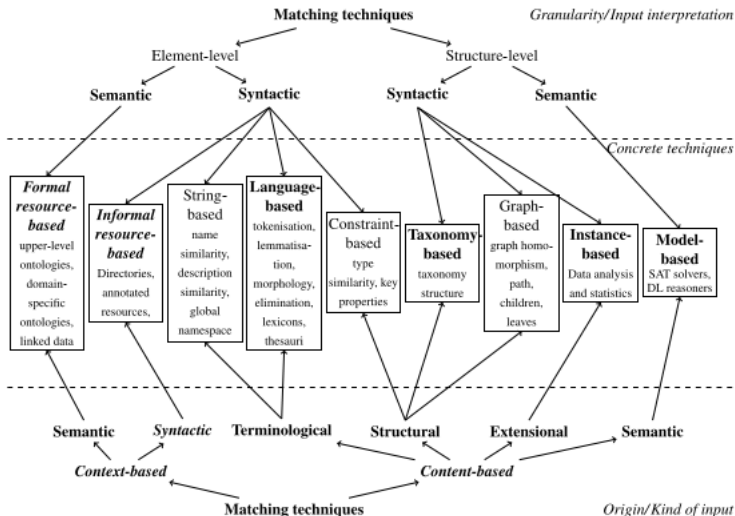
Ontology Alignment Evaluation Initiative (OAEI)

Classification of matching approaches

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Jérôme Euzenat, Pavel Shvaiko: *Ontology Matching*, Second Edition. Springer 2013, ISBN 978-3-642-38720-3, pp. I-XVII, 1-511

- Matching systems rely not on a single technique :
 - Sequential composition of alignments
 - Parallel composition of alignments
- Variety of approaches
 - Negotiation (dialogue)
 - Variations of argumentation frameworks
 - Consensus, voting
 - Intersection (maximising confidence)
 - Merge with filtering out the logical inconsistencies
 - Embeddings, LLMs,

Thank you !

Questions ?



Euzenat, J. and Shvaiko, P. (2013).

Ontology Matching, Second edition.

Springer Berlin Heidelberg, Berlin, Heidelberg.